LEARNING — the journey, Monday 12 to Saturday 24 October 2026

A report of understanding on the work of Andre van Zuydam and Paul Everitt in `mellum-worker`, written so Paul can follow how Andre's outcomes were reached. It explains how the easy and medium suites were cracked and how far the hard suite got. It covers every change since Monday 12 October, day by day, with what helped, what hurt and what the records do and do not support. It follows Paul's method so it can feed `learn/` directly.

The window is 288 commits: 145 by Paul and 123 by Andre (including merges). "This week" means Monday 19 to Saturday 24 October. The week before is a prologue (Part II).

How this report follows Paul's method

Paul's rule (source)	How this report applies it
Every step carries an evidence badge (evidence.md)	Every claim is tagged `reference-run`, `cited`, `screen` or `conjecture`. **No record** marks a screen known only from a commit message or LESSONS.md.
Screens find a lead; preregistered runs decide (sdd.md)	Nothing from this week is promoted. Every result Andre produced from 22 to 24 Oct is a `screen`, and Part VIII lists the preregistered run each one owes.
Research notes say what is measured and what is conjecture, and end with what would falsify the claim	Each crack in Part I ends with an Interpretation that names its weakest badge and the run that would upgrade it.
A lesson has six steps: Problem, Eval case, Evidence of the problem, Solution (with code), Validation, Interpretation (lesson-template.md)	Part I writes up easy, medium and hard in exactly those six steps, so each can become a lesson.
Compare inside a suite, never across suites. A missing metric is –, never `0`	The before/after tables follow this.
Before blaming the model, reconstruct its task, files, tool replies and environment, and check the scoring can tell the arms apart	This is the main finding (Part I and §IV.2): most failures were the harness.
Measure what ships; a setting the product does not share is a harness finding	Flow is a harness, not the shipped worker. Its results are harness findings, labelled as such.
Pathology → remedy (the ditch, the LC.7A Pathologies page)	Part IV is the pathology → remedy ledger, ready for `learn/pathologies.md`.
Reversals are recorded beside the original, dated	Part V lists every reversal and re-attribution.

Part I — How easy, medium and (most of) hard were cracked

The scoreboard

Suite (hidden grader tests)	Baseline Pi	Baseline Worker	Harness on, hidden tests passed	Flow + examiner, hidden
easy (4)	0/3 sessions	1/3	4/4 (stamped FAIL by a grading bug)	PASS
medium (18)	0/3	2/3	12/18	PASS
hard (34)	0/3	0/3	21/34	**29/34**; all 7 phases pass their own gates
Badge	`screen` (LC.11A, n=3)	`screen` (LC.11A, n=3)	`screen` n=1; arm config not recorded (LESSONS 22 Oct, `eeeae57`)	`screen` n=1 (`48010bc`; 24 Oct note)

The columns use different units, so read down a column, not across. Sessions passed is not the same unit as hidden tests passed. This follows Paul's rule to compare inside a suite.

The one fact that explains all three suites

The suite bases ship **no visible tests**. Look at `base/` in `learn/code/suites/*`: there is only `AGENTS.md`, `specs/` and `pyproject.toml`. The roadmap's `Tests:` lines ask the worker to *write* them. So in every suite the worker writes its own exam and grades itself on it. The hidden grader is the only external check.

That is rung 6's problem (AgentClinic phase 1 ships no test). Both Mellum and Flint fail rung 6 in direct mode. Flint is the second LLM Andre ran against Mellum; both score 11/12 on rungs 1–12, `screen`, no record.

Every crack below is one of three moves:

1. **Fix the check.** The harness was scoring a correct app as wrong.
2. **Instil the habit.** Prompt lines that stop the worker grading itself on a subset.
3. **Supply an external target.** An examiner writes the acceptance test from the spec before any code exists. The test is frozen so the coder cannot weaken it, and it accumulates across phases.

I.1 Easy — "it was already passing; the check was wrong"

Problem

Badge: `screen`

Baseline Pi fails easy 0/3 in about 22 s per session. The Baseline Worker passes 1/3. Easy is one prompt: "Implement Phase 1 — Home Page". It needs a shared Jinja2 layout, an HTML5 `lang="en"` page, a viewport, Bootstrap, a navbar and an exact tagline.

Eval case

Badge: `cited`

`uv run learn/code/harness.py run --suite easy --grade`. The hidden grader `learn/code/suites/easy/grader/test_phase1_home.py` has 4 tests (doctype, `lang`, 200 + tagline, nav links), plus the static gate in `grade()`.

Evidence of the problem

Badge: `screen`

LC.11A, n=3 per arm, llama.cpp:

Arm	easy passes
Baseline Pi	0/3
Baseline Worker	1/3
no-prompt	1/3
no-nudge	3/3
no-loop	2/3

On 22 Oct Andre's harness-on run passed **all 4 hidden tests and was still stamped FAIL**.

Solution

Badge: `cited` (commits with fail-first tests)

The failure was in the grader, not the worker. There were two bugs, fixed on Thursday 22 Oct:

- `5004187`: `harness.py` defined `grade()` twice. Python binds the second definition, so a strict all-rules ruff copy shadowed the intended `F,E9` gate. That failed working code on style nits (RUF100, I001).
- `af3b8b9`: ruff's `--exclude` *replaces* its default excludes, so `.venv` was linted. The worker runs `uv`, which creates a `.venv`, so every working solution failed on F403 inside PyYAML.

The code

```
# learn/code/harness.py: grade() lints only the worker's code
["ruff", "check", "--select", "F,E9", "--extend-exclude", "tests,grader_tests", ...]
```

Two habit remedies also apply to easy: the worker prompt v8 to v11 (run the whole file, the spec is the contract, self-check before finishing) and the suite fact `Do not: start servers`.

Validation

Badge: `screen`

- After the grade fixes, harness-on easy passes the hidden grader and the gate (n=1).
- Flow passes easy, visible and hidden (`48010bc`, n unstated).
- Regression tests: `grade()` tolerates a style-only issue; `harness.py` has no shadowed top-level definitions; a `.venv` with a star-import does not fail the suite. All three were shown to fail first.

Suite	Arm	Passes	Hidden tests	Source
easy	Baseline Worker (LC.11A)	1/3	—	`screen`
easy	harness on, before grade fix	stamped FAIL	4/4	`screen` n=1
easy	harness on, after grade fix	PASS	4/4	`screen` n=1
easy	flow	PASS	4/4	`screen` n unstated

Interpretation

The weakest badge is `screen` (n=1). Easy was cracked mostly by fixing the measurement.

Implication for Paul: the LC.11A easy numbers came from the harness *before* these fixes. Some Baseline Worker and no-* "fails" on easy may be grading artefacts. Before LC.11A's easy row is taught, re-grade the retained LC.11A workspaces with the fixed `grade()` (a cheap check, no model time), or rerun.

I.2 Medium — "the habit, then an examiner that asserts identity and redirects"

Problem

Badge: `screen`

Medium is four prompts (phases 1–4: home, board, add, resolve/reopen). The wall is phase 4. The spec says verbatim to add a stable `id: int` with an automatic, unique, never-reused default. The worker resolves and reopens by *list index* instead. That is the same id miss as ladder rung 9, and the A21 failure.

Eval case

Badge: `cited`

`--suite medium`. The hidden grader has 18 tests across `test_phase1_home`, `test_phase2_board`, `test_phase3_add` and `test_phase4_resolve_reopen`.

Evidence of the problem

Badge: `screen`

- LC.11A: Baseline Pi 0/3, Baseline Worker 2/3.
- Harness on: 12/18 hidden tests (n=1).
- A flow run went visible PASS / hidden FAIL because `Complaint` had no id field (`15c2766`, n=1).
- A later flow run looped to exhaustion on phase 3 (Add Complaint) against a gate no app could pass (`b4639c9`).

Solution

Badge: `screen`

Three layers, in the order they landed:

1. **Habit (prompt)**, Wed 21 Oct. v9: "The spec is the full contract. Implement EVERY requirement… unique ids that are never reused… even when no visible test checks it." v11: "Before you finish, re-read the spec… check your code against EACH clause." Answers *self-narrowing on the spec*.
2. **Control outside the model**, Thu 22 – Fri 23 Oct. `mellum-flow`:
   - plan → review_plan → examine → code → review_code → test, each phase an isolated Pi sub-agent;
   - only the coder writes;
   - the examiner writes the acceptance test from the spec;
   - the gate is frozen in `plan/gate/phase_<n>.py` and restored before scoring.
3. **Examiner rules** that close medium's two gaps:
   - Fri 23 Oct `15c2766`, *identity*: a phase that acts on a specific record must assert a stable explicit id field, distinct per record, addressed by id, unchanged across mutation.
   - Sat 24 Oct `b4639c9`, then `a1898ae`, *redirects*: TestClient follows redirects by default, so a correct 303 handler showed 200. The fix is to build the client once with `follow_redirects=False`.

The code

```
# examiner rule (mellum-flow.ts, paraphrased)
For any phase that resolves/reopens/updates/deletes/fetches a record: assert a stable
explicit id FIELD on the model — distinct per record, the route addressed by id, the
id unchanged across the mutation. Build the client as TestClient(app, follow_redirects=False).
```

Validation

Badge: `screen`

Suite	Arm	Passes	Hidden tests	Source
medium	Baseline Pi	0/3	—	LC.11A
medium	Baseline Worker	2/3	—	LC.11A
medium	harness on	—	12/18	`eeeae57`, n=1
medium	flow, before identity rule	visible PASS, hidden FAIL	—	`15c2766`, n=1
medium	flow	PASS	18/18	`48010bc`, n unstated

The phase-4 gate against the known-correct app went from 2 failures to 7/7 once the client-level redirect rule was in (`a1898ae`).

Interpretation

The weakest badge is `screen`: no n=3 flow run on medium, and the "medium PASS" in `48010bc` predates the identity and redirect rules.

What cracked medium is the combination of v9/v11 (tell it and make it check) and an external, frozen, identity-asserting gate. Which part is load-bearing is not separated. That needs an ablation: flow with and without the identity rule, n=3.

I.3 Hard — "all seven phases built; the gate is weaker than the oracle"

Problem

Badge: `screen`

Hard is 11 prompts: seven phases plus sqlite-utils migrations over one growing app, with an archive & audit phase, a `/tickets` report and a `/visits` report. Under the earlier harness Paul expected it to stay above the ceiling, and it did: 0/9 in every LC.11A arm.

Eval case

Badge: `cited`

`--suite hard`. The hidden grader has 34 tests across phases 1–6, `test_visits` and `test_migrations`.

Evidence of the problem

Badge: `screen`

- LC.11A: 0/3 Baseline Pi, 0/3 Baseline Worker.
- Harness on: 21/34 (n=1).
- In-session gates (plan-before-edit plus surgical) took hard to **0** (n=1).
- Early flow reached phase 4/7 and then appeared to stall.

Solution

Badge: `screen`

The harness remedies, in the order they were needed:

#	Day	Blocker	Remedy	Commit
1	Thu 22	A weak model waits out a gate inside its own session	Move control outside the model: an outer loop, then phases as tools	`cc758d1`, `48010bc`
2	Thu 22	Hard's spec is a 5.5k-character wall that drowns the model	Multi-level decomposition, plus a north star, running summary and last-phase context	`3d44ee1`
3	Fri 23	The "stall": oMLX's parser silently drops tool calls with large JSON arguments	Argument-free phase tools; content goes through `TASK.md`, `TODO.md`, `REVIEW.md`	`3bf2bb8`

#	Day	Blocker	Remedy	Commit
4	Fri 23	One attempt per phase is a coin flip	Best-of-N with a temperature spread (0.3–1.2); promote the first pass	`3bf2bb8`
5	Fri 23	Worker grades itself: visible 7/7, hidden 3/7	**Examiner**: an independent spec-only acceptance test before coding, frozen	`ae85226`
6	Fri 23	A later phase silently breaks an earlier one	Gates **accumulate**; every earlier gate is re-run	`fad46bb`
7	Fri 23	Phases 3–7 "passed" with no gate (best-of-N chose the empty fork)	A phase passes only with a real, compiling phase-N gate	`433157e`
8	Fri 23	Thrash: examine 5×, review 3×, never test	examine once per phase; 16-call cap	`632477d`
9	Fri 23	Planner empty with Pi's base prompt removed	Revert to `--append-system-prompt`; max_tokens 8,192	`48ed09e`, `b43d932`
10	Fri 23	Plan↔review infinite loop	Reviews fail open	`ccc049f`
11	Sat 24	Examiner gates no correct app could pass (redirects, lone fences, `"Resolve" not in html` while requiring "Resolved")	Client-level `follow_redirects=False`; strip fences; target the control	`a1898ae`, `6087b95`, `fadbb60`
12	Sat 24	Correct build failed on the coder's own scratch tests	Judge on the frozen gate only	`fadbb60`
13	Sat 24	Examiner emits nothing (thinking eats 16,384 tokens; then over-reasons a long prompt)	max_tokens 32,000; compress the prompt ~4,650 → ~2,560 characters	`da0b783`, `2196c75`

The code

```
# flow_run.py — a phase passes only on its own frozen gate
phase_gate_ok(n): plan/gate/phase_<n>.py exists, compile()s, defines ≥1 test
restore_gate():  copy every phase_1..n gate into tests/test_gate_phase_*.py, then pytest them
```

Validation

Badge: `screen` (n=1)

Suite	Arm	Phases passing own gate	Hidden tests	Source
hard	Baseline Worker	—	0/3 sessions	LC.11A
hard	harness on	—	21/34	`eeeae57`
hard	in-session gates	—	0	`c71068a`
hard	flow, pre-examiner	7/7	3/7 phases	LESSONS 23 Oct
hard	flow + examiner (Sat 24)	**7/7**	**29/34**	24 Oct note

All 5 remaining failures are in phases 6–7:

- `/tickets` sums are wrong, and "archived" is derived from status instead of `archived_at IS NOT NULL`.
- `/visits` returns an empty list.
- The migrations are not all recorded.

Phases 1–5 pass the hidden grader and were spot-checked by hand.

Interpretation

The weakest badge is `screen` (n=1, oMLX, concurrency 1). Hard is **not cracked**. What is shown is that the model can build all seven phases when the harness supplies decomposition, an external frozen target and bounded sequencing. The 29/34 is the **hidden-oracle gap**: the examiner's phase-6 and phase-7 gates accept an app the hidden grader rejects.

Two rules landed after the run and were **not re-run**:

- `5ad7c03`: computed endpoints are asserted with an included *and* an excluded row; migrations must be recorded.
- `6d88065`: self-contained tests, reuse of earlier gates' conventions, deltas instead of totals.

Validity caution for Paul: Andre tuned the examiner against hard while the hidden-grader failures were visible to him. The `archived_at` miss became the general rule "a definition keyed off a specific column". No suite names leak

(scrubbed in `2378022`), but the dev set and the test set are the same.

To upgrade:

1. A full hard re-run.
2. A preregistered `--mode flow` run on rungs 6–12, n=3.
3. A **held-out** suite the examiner was never tuned on.

Part II — Prologue: the week of Monday 12 October

Monday 12 — `/mellum last` and v0.1.7 (Paul)

- `/mellum last` shows the newest run record as a Markdown table with no model call. It replaced a slow "analyse my last run" prompt. Spec: `2026-10-12-mellum-last-design.md`, explicitly not a measured change.
- Review fixes: cells survive backslash-pipes, records are ordered by time.
- Release **v0.1.7**.

Tuesday 13 — A21 decides; the roadmap is triaged (Paul)

- **A21, preregistered.** The lighter `## Mellum` section plus a one-file roadmap scored **9/12**; the control also scored **9/12**. There were 0 outside writes. All four rung-9 failures were the same phase-4 id miss, and **every failing run passed its own test command**. The lighter section ships in **v0.1.8** (`cited`).
- **Rung-9 screen**, one change at a time, n=3: 1/3 and 1/3. Id misses were 5/6 with the one-file roadmap vs 1/6 with phase files. A lead, parked behind M.11 (`screen`).
- **Fix:** the ladder stops processes a run leaves behind and records them. **Fix:** back-to-back `/mellum` one-shots no longer record an empty task.
- **Roadmap triage:** instrument-only phases were retired, and rejected research code was removed (chunked and brief modes, strip-thinking, budget-hook, nine wording-ratchet tests). "No more instrument-only phases."

Sunday 18 — "don't trust self-exit" (Andre; PRs #1 and #2)

- **mlx-lm backend**, `--model-id`, sampling overrides.
- **Worker-exit lead**, recorded in AGENTS.md and skill §6. A self-exit is safe only when the oracle is external, complete and unweakenable. Verify by re-running the test, never by the worker's final message.
- **Exit-gate probes** (n=1, by hand): a one-shot misses the id; a frozen test plus re-dispatch converges at turn 25. "The oracle is necessary, not sufficient. The gate is load-bearing."
- **The mediated path in the ladder:** `--acceptance-test`, `--redispatch`, temperature and seed ramp, `--effort-ramp`, re-dispatch on timeout, the real failure in the retry message.
- **Mediated rung-9 screen:** stock FAIL at the 600 s deadline; mediated **PASS** in 194 s (n=1, `screen`).
- **Preregistered mediated-vs-stock run, rungs 6–9:** the stock arm scored **5/12** (deadline hits on a slow backend). Arm B **was never run**: there is no oMLX on Andre's machine.
- **Harness bugs:** an acceptance test named `acc_p4.py` was never collected; the worker blocked on `python app.py`; greedy retries repeated the same decode; a temperature ramp to 1.0 thrashed a working app (capped at 0.5).
- **Process:** mellum-dev skill, strawberry marker, CI for both suites, coding principles, "run tests visibly".

Part III — This week, day by day

Monday 19 October — "small models write tool calls as text"

- `content-tool-call-assist` (Andre). Promotes JSON or Hermes-XML text into real tool calls, gated on real tool names, for tiny models (Gemma E2B, Qwen 1.5B). Comes with the `patch-pi.sh` installer.
- **Tool-presentation experiment** (Gemma E2B, rungs 1–5, n=15): baseline **5/15**, stripped native tool field **0/15**. Reverted. The native tool block also renders tool results, so the ugly native format is load-bearing. Paul called it "the

cleanest experiment" (`screen`).

- **Prompt v6:** "Keep every existing test… add alongside." The commit itself says "no measured lift, no regression".
- **LESSONS.md** opens with 100 lessons. Skill §10: one small PR per win or learning.

Tuesday 20 October — "rung 9 falls; one Pi worker; the Learning Center is built"

Andre: v7 and the mediated stack.

- Prompt **v7** turns the worker into "make one failing test pass". It reads the test and its spec, creates the imported module, runs only the named test, never edits it and starts no servers.
- **Mediated 6–9: 12/12**, with rung 9 at **3/3** (`cited`). "A method stack, not an ablation."
- **v7 ablation, rung 9, n=3 per arm:**

Arm	Pass	Average wall
v6 prompt	3/3	701 s
whole-dir test command	3/3	**3,075 s**
grep/find restored	3/3	348 s
fixed high budget	3/3	**163 s**
(v7 win, reference)	3/3	272 s

The lever is the frozen test plus budget. Prompt and tools are cost knobs (`cited`).

- Tooling: `watch_ladder.py` (live trace follower), `compare_runs.py`.

Paul: LC.1 to LC.11, the Learning Center, in one day.

- LC.1: the site skeleton. LC.2: base Pi on **llama.cpp**, with `scripts/backend.py` replacing a Justfile made the same morning.
- LC.3: mini harness. LC.3A: suites and grading. LC.3B: the ladder takes a backend.
- **LC.3C, the preregistered llama.cpp reference run: 14/15** on rungs 1–5. Rung 4 repeat 3 stopped on `length` (`reference-run`; raw tarball still owed).
- LC.4: findings digest, ten lessons. LC.6: sessions and `--repeat`.
- **LC.7, the ditch:** a fabricated `Ran 4 tests… OK` over a real `FAILED (errors=4)`, and a dropped date filter (n=1). This is the pathology catalogue's origin.
- **LC.8 / LC.8A, one worker:** Mellum Mode became Mellum Worker; the pi-subagents Worker, intercom, delegated mode and OpenCode were removed.
- LC.9: worker-extension lesson. LC.10 and LC.11: two `cited` findings lessons.

Wednesday 21 October — "self-narrowing, two forms; gates don't add capability"

Mode and the frozen test.

- Shipped Mode with v7 on rung 9 scored **0/3**: v7 looks for a failing test that Mode never places.
- `--acceptance-test` in Mode plus `mellum-ncm` scored **3/3**, with NCM firing **0 times**. Credit goes to the test by name, not to "mediation" (`cited`).

Gates and model size.

- `mellum-ncm` (Andre): a next-concept verify gate that re-runs the test at settle and names the failing concepts.
- **3-bit halving: 3/8** (`cited`, a negative result). Quantization damage shows up as lost edit precision.
- **NCM × size:** 6-bit 3/3 with 0 fires; 3-bit and 1.5B fail, the 3-bit with "dozens" of fires. **"The gate makes a worker try, not able"** (`cited`; arms confounded, Mode vs direct).

Rungs 10–12. Phase 5 (Archive & Audit), phase 6 (reporting over a database) and phase 7 (summing columns of a context-overflowing file): **9/9** mediated, 7 of 9 runs re-dispatched (`cited`).

Observability, then a burst of extensions (52 minutes, 11:29–12:21).

- Feed, replay, control and the dashboard.
- `mellum-context` (FTS5 retrieval), `mellum-rethink` (push back on false human claims), `mellum-files` (PDF, Word, Excel), `mellum-vision` (local VLM), `mellum-override` (live prompt edits).

- All unit-tested, none ladder-measured (`conjecture`). Logged as THEORIES T1–T6.

The prompt sequence, v8 to v11, 14:11 to 16:23.

- Watching bare rungs 4 and 5 live through `mellum-override` showed **self-narrowing**: the worker narrows pytest to one `::test_fn`, passes it, and stops.
  - **v8**: run the whole file, pass every test. Bare ladder 6/12 → 11/12 claimed.
- Rung 9 still left `id=0`.
  - **v9**: "the spec is the full contract".
- The remaining failure was a foreground `uvicorn` blocking until the deadline.
  - **v10**: never start a server, plus the **foreground-server guard**.
- A 1-in-3 slip remained.
  - **v11**: re-read the spec clause by clause. Rung 9 **3/3 un-mediated**.
- All `screen`, **no record**.

Paul built the AgentClinic easy, medium and hard suites:

- known-good and known-broken qualification;
- a canary marking any session that touched the grader as contaminated;
- runs moved out of the repo;
- hard rebuilt to seven phases with sqlite-utils migrations;
- harness arms and evidence tables.

He also digested Andre's PRs #3 and #4. He marked v8–v11 and the 12/12 `screen` until reproduced on `main`, and the new extensions `conjecture`.

Thursday 22 October — "in-session gates regress; control moves outside"

Paul: LC.11A, the 45-session screen.

Arm	easy	medium	hard
Baseline Pi	0/3	0/3	0/3
Baseline Worker	1/3	2/3	0/3
no-prompt	1/3	2/3	0/3
no-nudge	3/3	2/3	0/3
no-loop	2/3	1/3	0/3

- Prompt: null. Loop breaker: null. Nudge: possibly harmful (a lead). Hard: no headroom.
- **Basic Remedies dropped**; the bundle became the **Baseline Worker**.
- The glossary is made automatic.

Andre, morning: fix the check.

- Andre's merge of `upstream/main` (`0367d04`) brought in Paul's work, but every file both had changed was resolved to Paul's side. That **discarded Andre's ladder runner changes**: `--acceptance-test`, `--redispatch`, `--effort-ramp`, `--pi-extension`, `--model-id`, `--temperature` and `--backend mlx-lm`, with their tests. It also discarded his 18–21 Oct changelog entries, the AGENTS.md run standard, three README sections, two backlog entries and `runs/` in `.gitignore`. Nothing is lost from history (`da04888` holds it all), but `main` has lacked it since (§VII.1).
- The shadowed `grade()` and `.venv` lint fixes cracked easy (I.1).
- LESSONS: "the harness as a habit formula, and its ceiling". Baseline for the suites: bare 0/3; harness on 4/4, 12/18, 21/34 hidden tests.

Andre, afternoon: gates inside the session.

- `mellum-quality` (ruff `F,E9` at finish), `mellum-plan` (plan-before-edit), `mellum-surgical` (anti-clobber), `mellum-contract` (route test derived from the spec).
- Wired together, these **regressed easy from 4/4 to 0 and hard from 21/34 to 0**. The plan gate deadlocked: 3,790 blocks, 0 edits. It was made fail-open and re-keyed on TODO.md.

Andre, evening: control outside the model.

- `mellum-loop` (outer state machine with plan and code review), then `mellum-flow` (phases as tools, isolated sub-agents) plus `mellum-nonblock`. **Easy and medium pass**, visible and hidden.
- Multi-level decomposition with a north star.
- `mellum-knowledge`: a BM25 lookup over the installed venv.

Friday 23 October — "the examiner: an external definition of done"

- **The stall**, a serving-format bug: argument-free tools, plus best-of-N.
- **Throughput:** 127 → 351 tok/s at N=1→16. A dependent chain gains nothing.
- **Examine phase** (`ae85226`) and **accumulating gates** (`fad46bb`).
- **Real gate required** (`433157e`). This exposed the morning's "7/7 visible": phases 3–7 had passed with no gate. **LESSONS.md carries no dated correction for this.**
- `confineToWorkspace` guard, after a worker ran `grep -rl` over `/Users`.
- `--mode flow` **in the ladder:** flow is scored by the rung's own hidden grader. Rung 6 goes FAIL → **PASS**.
- **Flint vs Mellum** direct: both 11/12, both fail only rung 6 (`screen`, no record).
- Thrash control: examine once, 16-call cap. Identity rule (cracks medium, I.2).
- **The replace/append reversal:** replacing Pi's base prompt left the planner empty, so it was reverted. Pi's base prompt is what makes this model emit a final answer.
- Reviews fail open, ending the plan↔review loop.

Saturday 24 October — "first full hard build; the hidden-oracle gap"

- **Nine examiner fixes**, each a gate no correct app could pass:
  - redirects;
  - lone fences;
  - substring absence;
  - scratch tests;
  - token budget;
  - prompt compression;
  - client-level redirects;
  - shared-state deltas;
  - completeness rules.
- **Hard: 7/7 phases visible, hidden 29/34** (I.3).
- Research note: the hidden-oracle gap.
- Andre added the Introduction and Terminology pages; terminology was then moved into the glossary.
- Windows and Linux install scripts, troubleshooting, backend version fixes.
- Merge of Paul's main (`f50ee9f`): CRLF converted back to LF so `backend.py`'s shebang works. `npm test` 492 pass, 1 fail; the `backend-lifecycle` kill test also fails on upstream.

Part IV — Pathology → remedy ledger

This is ready material for LC.7A (`learn/pathologies.md`).

IV.1 Worker pathologies

#	Pathology	Shown by	Remedy (artifact)	Badge
P1	False success, fabricated evidence	ditch (LC.7)	Re-run the test yourself; an external frozen acceptance test	`screen`
P2	Dropped requirement: the automatic id	A21 rung 9; medium phase 4	Frozen test (`--acceptance-test`); v9; v11; examiner identity rule	`cited` (12/12) / `screen`
P3	Self-narrowing on the tests (`::test_fn`)	bare rungs 4–5	Prompt v8	`screen`, no record

#	Pathology	Shown by	Remedy (artifact)	Badge
P4	Self-narrowing on the spec	rung 9 under v8	v9 + v11	`screen`, no record
P5	Foreground server blocks to the deadline	rung 9, 18 Oct screen	v10; foreground-server guard; `mellum-nonblock`; `Do not: start servers`	`screen`
P6	Test-command thrash, bent project	A18–A20	`## Mellum` facts	`cited`
P7	Weakened or deleted tests	rung 2 under v5	v6, v7 "never edit the named test"; frozen gates	`screen`
P8	Tool calls written as text	Gemma E2B, Qwen 1.5B	`content-tool-call-assist`	`screen`
P9	Whole-file clobber (942 → 2)	LESSONS	`mellum-surgical`	`conjecture` (0 clobbers in 82 traced edits)
P10	Machine-wide crawl	23 Oct flow run	`confineToWorkspace`	`conjecture`
P11	Real Python errors left behind (F811)	the shadowed `grade()`	`mellum-quality`	`conjecture`
P12	No target (no visible test)	rung 6; all three suites	Flow examiner	`screen`
P13	Grades itself on a weak gate	hard visible 7/7 vs hidden 3/7	Examiner plus accumulating gates; completeness rules	`screen`
P14	Orchestration thrash	rung 6 flow	Examine once; call cap	`conjecture`
P15	Output cap truncates a thinking model	rungs 4–5 at 4,096; plan at 2,048; examiner at 16,384	`max_tokens` must cover thinking plus output	`screen`
P16	Over-reasoning on a long prompt	examiner at 32,000	Keep the prompt lean (compression)	`screen`
P17	An incapable model under a gate loops	3-bit, 1.5B	None; add capability, not gates	`cited`
P18	A weak model waits out an in-session gate	plan gate: 3,790 blocks	Control outside the model (loop, flow)	`screen`

IV.2 Harness pathologies ("suspect the check first")

Defect	What it did	Fix
Shadowed `grade()`; ruff linting `.venv`	Failed a 4/4 hidden easy pass	`5004187`, `af3b8b9`
`run_suite` never copied `base/`	Suites ran in an empty workspace	`aa81d44` (caught by a smoke)
Recorder loaded before the worker	**Every arm ran as base Pi**	`5331417` (caught before the screen)
Acceptance test named `acc_p4.py`	Never collected	`ff17d60`
Greedy re-dispatch	Same decode each time	Seed and temperature ramp (capped at 0.5)
Retry message	Warning noise, no assert	`98f9d86`
oMLX drops large tool arguments	"Stall"	Argument-free tools (`3bf2bb8`)
Pass without a gate; best-of-N selected it	False phase passes	`433157e`
Write-list counted harness files	Correct app FAILed	bookkeeping under `plan/`
Gate follows redirects	Correct 303 FAILed	`a1898ae`
Substring absence check	No app could pass	`fadbb60`
Lone markdown fence	Gate would not compile	`6087b95`
Invented exact strings	Even the home page failed	`7954d85`
Scored on scratch tests	Gate-passing build FAILed	`fadbb60`
Fail-closed review	Infinite loop	`ccc049f`
Unread stderr pipe	Child hung for the whole timeout	drain stderr
`killpg` on a recycled pid	Best-of-N crash	`74efb7d` (`run_orient` still unguarded, `flow_run.py:162,166`)

Part V — Extensions and prompt changes, chronologically

V.1 Extensions

Added	Extension	Why (from the commit)	Badge	Lines
Mon 12, Paul	`mellum-last`	Model-free run table	not a measured change	168

Added	Extension	Why (from the commit)	Badge	Lines
Mon 19, Andre	`content-tool-call-assist`	Tiny models write tool calls as text	`screen`	450 (**no tests**)
Tue 20, Paul	`mellum-worker` (renamed from mode)	One vehicle; nothing Mellum receives changes	`cited`	620
Wed 21 02:07	`mellum-ncm`	Verify and re-dispatch inside any Pi session	`cited`: 0 fires	95
Wed 21 11:29	`mellum-feed`, `-replay`, `-control`, dashboard	Watch and steer a live worker	`conjecture`	113 / 90 / 51
Wed 21 12:01	`mellum-context`	Retrieve, don't load, large files	`conjecture` (T1–T4)	229
Wed 21 12:06	`mellum-rethink`	"The human can be wrong, or lying"	`conjecture` (T5)	179
Wed 21 12:10	`mellum-files`	Read PDF, Word, Excel	`conjecture` (T6)	257
Wed 21 12:16	`mellum-vision`	A local VLM for screenshots	`conjecture` (T6)	142
Wed 21 12:21	`mellum-override`	Live prompt edits; **found v8**	`conjecture`	91
Wed 21 15:37	foreground-server guard	Rung 9's blocking `uvicorn`	`screen` (with v11)	in guards
Thu 22 16:29	`mellum-quality`	Real errors at finish (F811)	`conjecture`	165
Thu 22 17:06	`mellum-plan`	Plan before the first edit	`screen`: **regressed**	118
Thu 22 17:13	`mellum-surgical`	Anti-clobber	`conjecture`	143
Thu 22 18:07	`mellum-contract`	Route test from the spec, no model	`conjecture`	132
Thu 22 18:29	`mellum-loop`	Control outside the session	`conjecture`	89
Thu 22 22:58	`mellum-flow` + `mellum-nonblock`	Phases as tools; servers threaded	`screen`: easy and medium pass; hard 29/34	531 / 88
Thu 22 23:56	`mellum-knowledge`	API lookup past the training cutoff	`conjecture` (0 calls observed)	74

The arc:

1. Capability and observability around one worker (Wed 21).
2. Gates inside the session, which regressed (Thu 22 afternoon).
3. Control outside the model (Thu 22 evening onward).

None of the 21 Oct burst is in the shipped product's measured path.

V.2 The worker prompt (`prompts/mellum-worker.md`, 607 → 2,326 characters)

Version	When	Added	Answered	Badge
v6	Mon 19	Keep every existing test	P7	"no measured lift, no regression"
v7	Tue 20	"Make one failing test pass": read the test and spec, create the module, run only the named test	test-chasing under v6	`cited` 12/12 mediated. **But** the ablation shows v6 also scores 3/3, and shipped Mode under v7 scores **0/3**
v8	Wed 21 14:11	Run the whole file; pass every test	P3	`screen`, no record
v9	Wed 21 15:14	"The spec is the full contract"	P2, P4	`screen`, no record. The commit says n=3; LESSONS and v11 say 2/3
v10	Wed 21 15:37	Never start a server; use TestClient in process	P5	`screen` (changed with the guard)
v11	Wed 21 16:23	Re-read the spec clause by clause before finishing (worked example: `Complaint(agent_name, text)` with a counter id)	the 1-in-3 slip	`screen` 3/3

What the sequence teaches:

- **Telling raises the hit rate; self-checking closes the miss.**
- Every line came from *reading what the worker actually ran*, which is Paul's rule. v8 was found by watching a live run through `mellum-override`.

Risks for Paul to weigh:

- v9 and v11 name rung 9's exact miss and model, so the prompt may teach to the test. A held-out rung would settle it.
- v10 and v11 each changed together with the guard.
- v11 ships, but Mode has never been measured under v11 without `--acceptance-test`.
- Every committed run record is v6 (3) or v7 (40).

V.3 Prompt lines outside the prompt file

- **Task line, Sun 18:** "Don't start servers". Paul flagged it as edging toward coaching.
- **Suite AGENTS.md** (all three suites):
  - `Test command: uv run --frozen python -m pytest tests`;
  - scoped changes;
  - run the test command and report;
  - `Do not: start servers`.

  Remedies for P5 and P6 are baked into the eval fixtures, so Baseline Pi with context files on already gets them.
- **Flow role prompts** (`mellum-flow.ts`, `flow_run.py`):
  - roles (22 Oct);
  - examiner (23 Oct 11:43);
  - strict negatives and edge cases (12:38);
  - identity (21:25);
  - no invented strings (21:59);
  - Pi base-prompt replace, then revert to append (21:59 → 23:02);
  - cohesive roles (23:15);
  - nine examiner rules (24 Oct).

  The examiner prompt went ~4,650 → ~2,560 characters and back up to ~3,770. These prompts live outside `prompts/`, the only prompt source (§VII).

Part VI — Code-level evaluation

Measured from `git log 4245ba3..HEAD`, excluding run records, lockfiles and fixtures.

- **Commit types.**
  - Andre: 44 `feat`, 38 `docs`, 14 `fix`, 8 `Flow:`, 18 untyped.
  - Paul: 76 `docs`, 37 `feat`, 12 `fix`, 9 `chore`, 7 `refactor`.
- **Code growth.**
  - extensions +4,267 / −578;
  - tests +2,951 / −1,190;
  - `learn/getting-started` +2,271;
  - scripts +1,238;
  - ladder code +878 / −599 (net shrink after the triage);
  - dashboard +765.
- **Hotspots.** `mellum-flow.ts` 24 commits, `run_ladder.py` 23, `harness.py` 22, `CHANGELOG.md` 123.
- **Fix chain.** The flow and examiner went through 27 commits in about 46 hours, nine of them `fix:` on Sat 24. That is fast learning, done in production against the suite being scored.
- **Test discipline.**
  - Paul: 1 of 42 code commits touched no test.
  - Andre: 15 of 63 since 19 Oct touched no test, including the stall fix, the append revert, fail-open reviews and the prompt compression.
  - `content-tool-call-assist` has none.
  - Totals now: 485 node tests and 238 Python tests. `docs-caps` 4/4.
- **Churn artefact.** The hard suite's `numbers.txt` (500,000 lines × 3) was added and then removed (`564c31e`), about 1.5M lines. `ladder/fixtures/agentclinic-p7/numbers.txt` (3.9 MB) is still tracked.
- **Clean:** no `trace.jsonl`, `stdout.txt` or `diff.patch` in the tree.
- **Debt.**
  - `run_orient killpg` unguarded.
  - The in-agent `test()` scores all of `tests/` while the runner scores only the gate.

- `visible` reflects only the last phase.
- `.idea/` and two ~3.5–3.9 MB PNGs committed.

Part VII — Fit with Paul's method: the audit

What already follows Paul's method, what does not, and the step that would bring each piece into line.

Conforms

- A21 and LC.3C were preregistered, with validity checks and disclosed reruns.
- The v7 ablation fixed its rule before the runs.
- LC.11A fixed its arms and n in a spec, screened first, and **dropped a module rather than overclaim**.
- Negative results are recorded as notes: tool strip, 3-bit, NCM × size.
- "Name the mechanisms" (`d547a63`) credits `--acceptance-test`, not "mediation".
- The 24 Oct note has a measured/conjecture header, a falsifier and risks.
- Examiner rules are stated generally, with suite names scrubbed.
- Fail-first tests on the grading and flow helpers.

VII.1 The 22 October merge: what `main` lost

Andre merged `upstream/main` into the fork on Thursday 22 Oct (`0367d04`). For every file both he and Paul had changed since their common base (`b7d987d`), the result equals Paul's side exactly. Andre's 18–21 Oct work is intact in history at `da04888`, an ancestor of `origin/main`. It is absent from the files on `main`:

Lost from `main`	Consequence today	Status (2026-10-24)
`ladder/ladder/run_ladder.py`, `pi_profile.py` and their tests: `--acceptance-test`, `--redispatch` (with the temperature and seed ramp), `--effort-ramp`, `--pi-extension`, `--model-spec`, `--model-id`, `--temperature`/`--top-p`, `--backend mlx-lm`, the focused test command and trimmed tools	The mediated path behind 12/12, the ablation, Mode 0/3 → 3/3 and rungs 10–12 cannot be rerun. `scripts/screen.sh`, `scripts/product.sh` and the dashboard's Start button (which runs `screen.sh`) call flags that no longer exist. `EXAMPLE.md` and `docs/backends/mlx-lm.md` describe them.	**Restored 2026-10-24** on the local branch `restore-mediated-ladder` (not pushed): hand-merged onto Paul's LC.3B flags, `--backend`/`--model-id`/`--model-spec` dropped in their favour; ladder suite 189 passed (158 on `main`)
`CHANGELOG.md [Unreleased]`: all of Andre's 18–21 Oct entries	The changelog did not describe what ships (v11, the extensions)	**Restored**: the 17 entries whose code survived, then the ladder entries rewritten for the restored code, under single headings
`AGENTS.md` run standard and "Driving the worker: don't trust its self-exit"	Undocumented, and the scripts it names were broken	**Restored**, updated for `--model omlx/<id>` and the archived M.11 row
`README.md`: "This session — 2026-10-18", "the strawberry skill", "Getting started (the next engineer)"	Lost onboarding text	**Two restored**. "This session" left out: it repeats claims Paul's digest marked unsupported ("A21 at 0/3", "~150× faster")
`BACKLOG.md`: "Freeze-the-oracle exit gate", "mlx_lm.server vs oMLX"	Two reopen conditions lost	**Restored**, the second narrowed to servers (sampling sits under Paul's "Model-parameter impact")
`.gitignore`: `runs/`	`screen.sh` output is no longer ignored	**Restored**

Paul's removals of the subagent and OpenCode templates in LC.8 and LC.8A were deliberate, and they stay removed. The extensions, prompt v11, rungs 10–12 and the foreground-server guard survived the merge.

VII.1a Does the lost ladder code fit Paul's method?

Mostly yes as an instrument; not yet as evidence. The code measures a question Paul himself kept open: his BACKLOG entry "The mediated test" names Andre's PR #1 as the candidate, and the archived M.11 row describes this test. Paul's own digests badge each piece within his method (PR digest, follow-up digest).

Piece	Paul's badge	Fits	Does not fit
`--acceptance-test` (place an external, frozen test)	`screen`	The exit-gate principle (external, complete, unweakenable); scored by Paul's own hidden tests and write list; lives in the ladder, not a scratch driver	Its first PASS was n = 1 and confounded with the no-servers line and an unused re-dispatch budget

Piece	Paul's badge	Fits	Does not fit
`--redispatch N` (re-run the frozen test, hand back the failure)	`screen`	Verifies the exit instead of trusting it	Three things changed between the two probes (freeze, re-dispatch, budget)
Temperature and seed ramp on retry	`conjecture`	Default off; existing records unaffected	Nothing shows a retry passing because of it; the kind of knob the 13 Oct triage cut
`--effort-ramp`	`screen` (deadline re-dispatch)	Spends compute in proportion to measured difficulty	The v7 ablation found fixed-high budget fastest (163 s), so the ramp is the wrong default on hard rungs
`--backend mlx-lm`, `--model-id`, `--temperature`/`--top-p`	runs on mlx-lm stay `screen`	Defaults leave oMLX runs unchanged	Overlaps Paul's LC.3B `--provider`, `--base-url`, `--model`, which do the same job
`--pi-extension`, `--model-spec`	—	Lets an extension such as `mellum-ncm` be measured without shipping it	—
`--acceptance-test` in `--mode mode` (`faaed7e`)	`screen`	Measures the shipped worker, as "measure what ships" asks: Mode 0/3 → 3/3	n = 3, recorded only on the fork

Against Paul's rules as a whole:

- **Conforms:**
  - The experiments came with screens first, research notes with honesty headers, and a written preregistration (mediated vs stock, rungs 6–9) before any deciding run.
  - Every new flag defaults off, so no measured setting changed.
  - Commits carried tests.
- **Does not conform:**
  - *No record on* `main`*.* Paul: entries stay screens "until a record lands here". Today the records are on the fork's `main` but the code that produced them is not, so they cannot be reproduced.
  - *The deciding run never ran.* Arm B of the preregistration needed oMLX, so no promotion was earned.
  - *The early screens ran a setup the product does not ship* (direct mode, mlx-lm, `tuned`, guards). M.11 asked for Mode. That makes them harness findings, not worker claims, until the Mode runs.
  - *The harness got thicker.* Two ramps and a second backend surface, against "keep the harness thin".

A restore that conforms:

1. Bring back the M.11 core: `--acceptance-test`, `--redispatch`, the Mode support and `--pi-extension`.
2. Express the backend choice through Paul's LC.3B flags rather than `--backend` and `--model-id`.
3. Keep the ramps opt-in, labelled `conjecture` in the changelog.
4. Then run arm B, or its llama.cpp equivalent, so the 12/12 can be reproduced on `main` and earn `cited`.

Done 2026-10-24 (steps 1–3), on the local branch `restore-mediated-ladder`*.* The restored tests were shown failing against `main`'s runner (29 failures), then passing on the merge. One change goes beyond a straight restore: Andre's code had trimmed the tool set to five for *every* direct run, which would have changed Paul's measured unmediated runs (the llama.cpp reference used seven). The trim now applies only when `--acceptance-test` is set, as Andre's own changelog described it, with a fail-first test. Not restored: `--model-spec` (small-model screens only; still at `da04888`). Step 4 is still owed.

VII.2 Does not conform yet, and the step that fixes it

Gap	Rule it breaks	Step to conform
Flow (22–24 Oct) has no roadmap phase, spec or plan; `## Now` reads "Active: none"	SDD cycle; `## Now` is a pointer	Add an LC row (or Andre-track rows) for flow and the examiner. Write a spec, since product behaviour changed in the ladder (`--mode flow`)
v8–v11 ship with no `docs/ladder/` record; v11 is unmeasured in Mode	Measured settings change only with a recorded ladder run	Preregister: v11 vs v7, Mode and direct, rungs 1–12, n=3. Record it. Revert to the last recorded prompt if it loses
Flow, Flint parity, bare 6/12, the 1.5B result: no records	Skill §11: record every run that backs a claim	Commit `table.md` + `run.json` under `docs/ladder/`; raw evidence in a `research-2026-10-2x` release
New defaults with no ladder run: `confineToWorkspace` on, suites easiest-first, per-phase `max_tokens`	Measured settings	Run one ladder arm with and without, or make each opt-in

Gap	Rule it breaks	Step to conform
Flow and examiner prompts live in code	`prompts/mellum-worker.md` is the only prompt source	Move the role prompts under `prompts/` (for example `prompts/flow/*.md`) or amend AGENTS.md by decision
The 22 Oct merge dropped Andre's ladder code and records (§VII.1)	Measure what ships; keep the changelog current	**Done 2026-10-24** on a local branch; reproducing the 12/12 on `main` (arm B) is still owed
Duplicate `Added`/`Fixed` headings; flow sub-agents said to "REPLACE" Pi's prompt; examine at 16,384; no entries for `48ed09e`, `ccc049f`, `3fc1d27`, `891721a`, the intro page or the install scripts	Entries say what changed	**Done 2026-10-24**: one heading each; the replace/revert history stated; 32,000 noted; entries added
`mellum-flow.ts` `BASE_GUIDE` comment described the reverted `--system-prompt`	Code reads true	**Done 2026-10-24**
README: "a 643-character worker prompt" (v5), "15/15"	README states what ships	Update once v11 has a record, or state plainly that v11 ships unmeasured
LESSONS 23 Oct "seven phases visible" without a correction after `433157e`; "Proven (measured)" on n=1 runs	Reversals recorded beside the original, dated	**Done 2026-10-24**: dated corrections beside both
v9 commit says 3/3; LESSONS says 2/3	Claims match the evidence	**Done 2026-10-24**: dated correction in LESSONS (2/3 stands; the screens have no record)
LC.11A easy rows graded by the pre-fix `grade()`	Before blaming the model, check the scoring	Re-grade the retained workspaces (no model time)
Examiner tuned on the suite it is scored on	Check the scoring can distinguish; screens are not rates	A held-out suite or rung before any lesson claims it
`research-2026-10-20` not released	A `reference-run` needs raw evidence	Publish the tarball
AGENTS.md cites OpenCode for the install-script rule	Accurate rationale	Re-word for Pi's git install
Site deploys from `main` before Andre's linkable consent (LC.5)	Spec gate	Post the consent quote
Intro page: "Andrew Markov" should be Andrey; framing of LLMs as grown N-grams	Accuracy	Review before readers rely on it

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Part VIII — Mapping to `learn/`: from a basic harness to an enhanced one

The learner's arc is the suite scoreboard in Part I. **Baseline Pi fails easy, medium and hard.** Each lesson adds one remedy for one pathology and shows the scoreboard move, in Paul's six steps. Getting Started is the basic harness. The enhanced lessons slot into Paul's LC rows where one fits, and otherwise into Andre's track (LC.5).

Part 1 — the basic harness (exists)

Lesson	Teaches	Status
GS 1 Base Pi on llama.cpp	backend, private profile	shipped
GS 2 The ditch	P1: false success, fabricated evidence, dropped requirement	shipped (n=1)
GS 3 Mini harness	one isolated, scored run	shipped
GS 4 Suites and telemetry	hidden graders, qualification, reading a trace	shipped; re-run owed on the rebuilt hard
GS 5 The worker extension	extension mechanics, `/mellum`	shipped
Lessons: Baseline	Baseline Pi 0/9 vs Baseline Worker 3/9	shipped (`screen`)
LC.7A Pathologies	the catalogue: **Part IV of this report**	not started

Part 2 — the enhanced harness (proposed order)

#	Lesson (pathology → remedy)	Artifact	Row	Badge now	Run owed
E1	See what it actually did	`watch_ladder.py`, feed, dashboard	Andre	`conjecture` (a how-to)	none
E2	Suspect the check first (**how easy was cracked**)	grade fixes; known-good qualification	LC.7A / Andre	`cited` (fail-first tests)	re-grade LC.11A easy

#	Lesson (pathology → remedy)	Artifact	Row	Badge now	Run owed
E3	Thinking is working memory: caps that truncate	per-role `max_tokens`	**LC.12**	`screen`	suite at two caps, n=3
E4	Servers block	v10, foreground-server guard, nonblock, suite fact	**LC.13**	`screen`	medium with and without the guard
E5	Project facts	`## Mellum`, suite AGENTS.md	**LC.14**	`cited`	llama.cpp reference run
E6	Write only what you list	surgical, `confineToWorkspace`, write-list, `plan/`	**LC.15**	`conjecture` / `cited`	suite with and without
E7	Self-narrowing, two forms	v8; v9 + v11	Andre	`screen`	preregistered v11 vs v7
E8	Don't trust the exit: freeze an external test	`--acceptance-test`, `--redispatch`	Andre (mediated-test backlog)	`cited` 12/12 + ablation	Mode arm B
E9	A gate makes it try, not able (negative)	`mellum-ncm`; 3-bit; 1.5B	Andre	`cited`	—
E10	Who holds the loop: in-session gates regress	plan (regression) → loop → flow	**LC.16**	`screen`	flow vs worker, n=3
E11	One spec file per phase: decomposition	`flow_run.py` split, north star	**LC.19**	`screen`	—
E12	The examiner, a spec-derived definition of done (**how medium was cracked**)	examine, frozen accumulating gates, identity rule	Andre	`screen`	`--mode flow`, rungs 6–12, n=3
E13	Writing tests a correct app can pass	examiner rules (redirects, substrings, deltas)	Andre	`screen`	gate vs known-good per phase
E14	The hidden-oracle gap (**where hard stands**)	completeness rules	Andre	`conjecture` (not re-run)	hard re-run, then a held-out suite
E15	Grounded knowledge past the cutoff	`mellum-knowledge`	Andre	`conjecture`	measure lookup use on hard
E16	Sidebar: tiny models and tool formats	assist; 0/15 strip	Andre	`screen`	—

Paul's rows with no Andre counterpart. LC.17 (a prose brief stops) and LC.18 (the intercom prefill tax) stand alone. LC.10 and LC.11 are written.

Before any Andre-track lesson publishes:

- The LC.5 consent quote.
- Steps 3 and 5 of each lesson backed by the run in "Run owed", recorded in `docs/ladder/`.
- The Part VII gaps closed, so lessons can cite `main`.

Part IX — Open questions and runs owed

0. Andre's ladder runner code is restored on the local branch `restore-mediated-ladder` (§VII.1a); merge it into the fork's `main` so items 1 and 2 can run there.
1. Mediated vs stock, rungs 6–9, arm B (preregistered Sun 18, never run).
2. v11, preregistered, in Mode and direct. Then a held-out rung for leakage.
3. Hard re-run with the 24 Oct examiner rules. Does 29/34 close, and does the ~3.8k-character examiner still converge on every phase?
4. `--mode flow` vs direct, and Flint vs Mellum, n≥3.
5. Recalibrate hard so arms separate under the Baseline Worker.
6. No-nudge 5/9 vs Worker 3/9 at a larger n.
7. Best-of-N (concurrency > 1) vs sequential retries.
8. Re-grade LC.11A easy with the fixed `grade()`.
9. Release `research-2026-10-20`. Commit the Sun 18 records. Re-baseline the 6-bit with guards on.

Sources

- **Research:**
  - `docs/superpowers/research/2026-10-13-*`

  - `docs/superpowers/research/2026-10-18-*`
  - `docs/superpowers/research/2026-10-19-*`
  - `docs/superpowers/research/2026-10-20-*`
  - `docs/superpowers/research/2026-10-21-*`
  - 2026-10-24 flow note
- **Records:**
  - `docs/ladder/ablation-a21-*`
  - `docs/ladder/screen-rung9-*`
  - `docs/ladder/2026-10-20-*`
  - `docs/ladder/2026-10-21-*`
  - `docs/ladder/reference-llamacpp-rungs1-5`
- **Specs:**
  - `2026-10-12-mellum-last`
  - `2026-10-20-learning-center`
  - `2026-10-20-one-pi-worker`
  - `2026-10-21-basic-remedies`
  - `2026-10-22-eval-suite-pipeline`
- **Plans:** `docs/superpowers/plans/archive/2026-10-20-*`, `…/2026-10-21-basic-remedies.md`
- **Root documents:**
  - LESSONS.md
  - THEORIES.md
  - CHANGELOG.md
  - ROADMAP.md
  - BACKLOG.md
  - AGENTS.md
  - README.md
  - docs/sdd.md
- **Method:**
  - evidence.md
  - lesson-template.md
  - ditch.md
  - lessons/baseline.md
- **Code:**
  - `git log -p -- prompts/mellum-worker.md`
  - `extensions/*.ts`
  - `learn/code/{harness,flow_run,loop_run,evidence}.py`
  - `learn/code/suites/*/{base,grader}`